

Fabric Structure and Design

1.Course Code:	FE 305-0723
2. Course Title:	Fabric Structure and Design.
3. Credit:	3.0
4. Rationale:	

To provide in-depth knowledge in the areas of woven fabric structure, loom selection for different woven fabric manufacturing, properties and their end uses.

5. Course Content:

Part A

Introduction: Classification of woven fabrics, Identification of warp and weft yarn, Definition related to structure and design, parts of a complete design, Methods of indicating drafting, Classification of drafting. Loom selection for specific design.

Plain Weave Fabrics: Features and classification of plain weave. Derivatives of plain weave and their characteristic, Ornamentation of plain weave fabrics. Commercial plain fabrics and uses.

Twill-weaves: Features and classification of twill weave. Derivatives of twill weaves and their characteristics. Influence of yarn twist on the twill lines. Angle of inclination of twill weaves. Commercial twill fabrics and their uses.

Satin and sateen weaves: Features of satin / sateen weaves, classification, and move number selection. Derivatives of satin weaves. Commercial satin fabrics and their uses.

Fancy designs of fabrics: Characteristic, Construction principle and uses of huckaback, mock leno, distorted thread effects, honeycomb, bedford cord, pique and crepe weaves, stripe and checks

Part B

Color & Weave effects: Introduction to colour and weave effect. Simple colour and weave effects (pinstripe, crow's foot, dog's tooth, shepherds check etc.) Compound colour and weave effects, Stripe and check effects

Figuring with extra threads: Characteristic, Construction principle and uses of Extra warp and extra weft design.

Double Cloth: Principle of double width, tubular, multiply fabric. Characteristic, classification, Construction principle and uses of stitched double cloth.

Commercial Fabrics: Specification, characteristics and their uses.

6. Course Learning Outcomes (CLOs) and Mapping of CLOs with Program Learning Outcomes (PLOs)

[illegible]

CLO2	Explain and illustrates the structures of woven fabric, their derivatives.	✓											
CLO3	Develop new design based on the basic structures.			✓									
CLO4	Decide whether it is possible or not to produce the woven fabric by the existing weaving loom.		✓										
CLO5	Identify different commercial fabrics produced in the local and export-oriented industries.	✓											

7. Course plan specifying content, CLOs, co-curricular activities (if any), and assessment strategy mapped with CLOs.

Lesson	Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLO
Part- A				
1	Introduction: Classification of woven fabrics, Identification of warp and weft yarn, Definition related to structure and design, Parts of a complete design, Methods of indicating drafts	Multimedia Presentation (MMP) and Whiteboard, Online platform, Video	Verbal interaction, sudden test, class test, Final examination.	CLO1
2	Introduction: Classification of drafting, Fabric weight calculation.	Multimedia Presentation (MMP) and Whiteboard, Online platform, Video	Verbal interaction, sudden test, class test, Final examination.	CLO1, CLO4
3	Plain Weave Fabrics: Features and classification of plain weave, Derivatives of plain weave and their characteristic	Multimedia Presentation (MMP) and Whiteboard, Online platform, Video	Verbal interaction, sudden test, class test, Final examination.	CLO1, CLO3
4	Plain Weave Fabrics: Ornamentation of plain weave fabrics, Commercial plain fabrics and uses, Advantages	Multimedia Presentation (MMP) and Whiteboard,	Verbal interaction, sudden test, class test, Final examination.	CLO3, CLO5

	and disadvantages of plain weave fabric.	Online platform, Video		
5	Twill- weaves: Features and classification of twill weave, Derivatives of twill weaves and their characteristics, Influence of yarn Twist on the twill lines. Angle of inclination of twill weaves	Multimedia Presentation (MMP) and Whiteboard, Online platform, Video	Verbal interaction, sudden test, class test, Final examination.	CLO1, CLO3
6	Twill- Weaves: Commercial twill fabrics and their uses, advantages and disadvantage of twill weave fabric.	Multimedia Presentation (MMP) and Whiteboard, Online platform, Video	Verbal interaction, sudden test, class test, Final examination.	CLO1, CLO5
7	Satin and Sateen weaves: Features of satin / sateen weaves, Classification and move number selection, Derivatives of satin weaves, commercial satin fabrics and their uses.	Multimedia Presentation (MMP) and Whiteboard, Online platform, Video	Verbal interaction, sudden test, class test, Final examination.	CLO2, CLO4
Part-B				
8	Fancy Design of Fabrics: Characteristic, Construction principle and uses of huckaback, mock leno	Multimedia Presentation (MMP) and Whiteboard, Online platform, Video	Verbal interaction, sudden test, class test, Final examination.	CLO1
9	Fancy Design of Fabrics: Distorted thread effects, Honeycomb, Bedford cord, pique and crepe weaves.	Multimedia Presentation (MMP) and Whiteboard	Verbal interaction, sudden test, class test, Final examination.	CLO2
10	Color & Weave Effects: Introduction to color and weave effect, Simple color and weave effects (pinstripe, crow's foot, dog's tooth, shepherds check etc.)	Multimedia Presentation (MMP) and Whiteboard, Video	Verbal interaction, sudden test, class test, Final examination.	CLO1
11	Color & Weave Effects: Compound color and weave effects (Stripe and checks effects).	Multimedia Presentation (MMP) and Whiteboard	Verbal interaction, sudden test, class test, Final examination.	CLO2
12	Figuring with extra threads	Multimedia Presentation (MMP) and Whiteboard	Verbal interaction, sudden test, class test, Final examination.	CLO1

	Characteristic, Construction principle and uses of Extra warp and extra weft design			
13	Double Cloth Principle of double width, Tubular, Multiply fabric, Characteristic, Classification, Construction principle and uses of stitched double cloth.	Multimedia Presentation (MMP) and Whiteboard, Video	Verbal interaction, sudden test, class test, Final examination.	CLO1, CLO3
14	Fabrics Specification, Characteristics and their uses.	Multimedia Presentation (MMP) and Whiteboard	Verbal interaction, sudden test, class test, Final examination.	CLO1

8. Assessment Plan:

Bloom's Category	CIE* (40%)			SEE** (60%)
	Attendance/ Performance (10%)	Class Test/ Quiz (20%)	Assignment/ Presentation (10%)	
Remember		25%		10%-20%
Understand	100%	50%		30%-40%
Apply		25%	50%	40%-60%
Analyse			50%	
Evaluate				
Create				

*CIE- Continuous Internal Evaluation

**SEE- Semester End Examination

9. Reference Materials

- Watson's Textile Design- Z. J. GROSICKI (Published by Woodhead Publishing Limited, Abington Hall, Abington, Cambridge CB1 6AH, England, 2004)
- Understanding Textiles for a Merchandiser- Prof. Dr. Engr. Shah Alimuzzaman Belal (Published by LB Graphics & Printing, 2016)
- Watson's Advanced Textile Design: Compound Woven Structures- Z. J. GROSICKI (Published by Woodhead Publishing Limited).

Fabric Structure and Design Lab

- 1.Course Code:** FE 306-0723
- 2. Course Title:** Fabric Structure and Design Lab.
- 3. Credit:** 1.0
- 4. Rationale:**

To deliver hands-on experience regarding the construction and design techniques of woven fabric so that the students can investigate the structural, aesthetic and functional behavior of sample based on fabric type and take an appropriate decision to reproduce the reference one.

5.Course Content:

General procedure for the analysis of woven fabric, Determination of specifications for the reproduction of fabric samples of all types. Calculations relevant to the production of different types of woven fabric.

6. Course Learning Outcomes (CLOs) and Mapping of CLOs with Program Learning Outcomes (PLOs)

No	CLO Statement	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1	Perform efficaciously as an individual or in a team to identify structural, constructional and aesthetic behavior of fabrics based on their weave type									✓			
CLO2	Select the basic weave structure and draw the pattern design correctly to develop the specified texture appearance on fabric surface			✓									
CLO3	Analyze the design and structure of specimens and choose suitable loom and weave plan to reproduce the reference one.		✓										
CLO4	Apply the skill of design analysis to develop a new one according to the requirement of buyers.				✓								

7.Course plan specifying content, CLOs, co-curricular activities (if any) and assessment strategy mapped with CLOs.

Lesson	Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLO
1	Introduction and of the course, lab equipment and a brief description about how to analyze fabric design and represent it on graph paper.	A brief Lecture on the experiment), Demonstration, provide	Verbal interaction, quiz, sudden test, class test, Final examination, Assignments	CLO2

		performance sheet, performing experiment by the students		
2	Determination of specifications for the reproduction of fabric samples, Calculations relevant to the production of different types of woven fabric.	A brief Lecture on the experiment), Demonstration, provide performance sheet, performing experiment by the students	Verbal interaction, quiz, sudden test, class test, Final examination, Assignments	CLO1, CLO2
3	Analysis of plain and derivatives of plain weave.	A brief Lecture on the experiment), Demonstration, provide performance sheet, performing experiment by the students	Verbal interaction, quiz, sudden test, class test, Final examination, Assignments	CLO1, CLO2, CLO3
4	Analysis of twill weaves.	A brief Lecture on the experiment), Demonstration, provide performance sheet, performing experiment by the students	Verbal interaction, quiz, sudden test, class test, Final examination, Assignments	CLO1, CLO2, CLO3
5	Analysis of derivatives of twill weave	A brief Lecture on the experiment), Demonstration, provide performance sheet, performing experiment by the students	Verbal interaction, quiz, sudden test, class test, Final examination, Assignments	CLO1, CLO2, CLO3
6	Analysis of derivatives of twill weave	A brief Lecture on the experiment), Demonstration, provide performance sheet, performing experiment by the students	Verbal interaction, quiz, sudden test, class test, Final examination, Assignments	CLO1, CLO2, CLO3

7	Analysis of derivatives of twill weave	A brief Lecture on the experiment), Demonstration, provide performance sheet, performing experiment by the students	Verbal interaction, quiz, sudden test, class test, Final examination, Assignments	CLO1,CLO2, CLO3
8	Analysis of satin/sateen weave.	A brief Lecture on the experiment), Demonstration, provide performance sheet, performing experiment by the students	Verbal interaction, quiz, sudden test, class test, Final examination, Assignments	CLO1,CLO2, CLO3
9	Analysis of fancy weaves.	A brief Lecture on the experiment), Demonstration, provide performance sheet, performing experiment by the students	Verbal interaction, quiz, sudden test, class test, Final examination, Assignments	CLO1,CLO2, CLO3
10	Analysis of fancy weaves.	A brief Lecture on the experiment), Demonstration, provide performance sheet, performing experiment by the students	Verbal interaction, quiz, sudden test, class test, Final examination, Assignments	CLO1,CLO2, CLO3
11	Analysis of fancy weaves.	A brief Lecture on the experiment), Demonstration, provide performance sheet, performing experiment by the students	Verbal interaction, quiz, sudden test, class test, Final examination, Assignments	CLO1,CLO2, CLO3
12	Analysis of compound structures (weave and color effect).	A brief Lecture on the experiment), Demonstration, provide	Verbal interaction, quiz, sudden test, class test, Final examination, Assignments	CLO1,CLO2, CLO3

		performance sheet, performing experiment by the students		
13	Analysis of extra warp design and extra weft design	A brief Lecture on the experiment), Demonstration, provide performance sheet, performing experiment by the students	Verbal interaction, quiz, sudden test, class test, Final examination, Assignments	CLO1,CLO2, CLO3
14	Analysis of commercial fabric	Provide performance sheet, performing experiment by the students	Verbal interaction, quiz, sudden test, class test, Final examination, Assignments	CLO4

8. Assessment Plan

Bloom's Category	CIE* (60%)			SEE** (40%)	
	Lab Attendance (10%)	Experiment and Performance (40%)	Lab Report (10%)	Experiment (20%)	Viva (20%)
Cognitive domain	10				20
Affective domain			10		
Psychomotor domain		40		20	

*CIE- Continuous Internal Evaluation

**SEE- Semester End Examination

9. Reference Materials

- Understanding Textiles for a Merchandiser by Engr. Shah Alimuzzaman Belal
- Watson's Textile Design and colour: Elementary weaves and Figured Fabrics by William Watson and Z Grosicki, seventh edition, Wood head Publishing; 2014.
- Watson's advanced textile design: Compound woven structures by Z Grosicki, fourth edition Woodhead Publishing, 2014
- Fabric structure and design by Gokarneshan, N. (2004), New Age International.
- Grammar of Textile Design by H. Nisbet (1919)
- A manual on Weave construction by Ivo Kastanek
- Technology of Textile Design by E.A. Posselt